Cinnamon Agronomy Knowledge Base: Guidelines for Fertilizer, Pests, Diseases, and Harvesting

Cinnamon Advisor LLM

August 19, 2026

Contents

1 Introduction

This document serves as the foundational knowledge base for the Cinnamon Fertilizer Advisor's Retrieval-Augmented Generation (RAG) pipeline. It aggregates critical agronomy data, primarily sourced from the Department of Cinnamon Development (DCD) Sri Lanka and global agricultural bodies, to provide the Large Language Model (LLM) with factual, up-to-date guidelines on cinnamon cultivation, fertilizer management, pest and disease control, and harvesting.

2 Environmental Requirements for Cinnamon Cultivation

Cinnamon (Cinnamomum verum J. Presl) is a tropical, evergreen perennial plant. While highly adaptable, commercial cultivation requires specific environmental conditions to optimize bark yield and volatile oil content.

2.1 Climatic Conditions

- **Temperature:** The optimal temperature range is between 25°C and 35°C.
- **Rainfall:** Commercial cultivation thrives in areas with an annual rainfall exceeding 1875 mm. In drier regions with rainfall below this threshold, cultivation is possible provided adequate irrigation is available.
- **Elevation:** Cinnamon can be commercially grown from sea level up to an elevation of 700 meters. Growth and yield tend to decline at higher elevations.
- **Humidity:** The optimal relative humidity is between 75% and 85%.

2.2 Soil Requirements

Cinnamon requires deep, well-drained soils.

- **Wet Zone:** Red-Yellow Podzolic soils, common in the rolling hills of the wet zone, are highly suitable. These soils are typically somewhat acidic.
- **Coastal Areas:** Flat coastal lands with sandy or sandy loam soils are excellent for cinnamon cultivation.
- **Unsuitable Conditions:** Avoid lands with a soil depth of less than 1 meter, rocky terrain, poor drainage, fluctuating water tables, or high salinity. Cinnamon roots are highly susceptible to root rot; prolonged waterlogging (over two weeks) will result in root death. Avoid lands with a slope exceeding 60%.

3 Fertilizer Management

Proper nutrient management is critical for maximizing the yield of cinnamon bark and leaves. The Cinnamon Fertilizer Advisor LLM must strictly adhere to the following guidelines when advising farmers.

3.1 Soil pH and Dolomite Application

- **Optimal pH:** The optimal soil pH for cinnamon is between 5.5 and 6.5.
- **Acidity Correction:** If the soil pH falls below 5.0, Dolomite must be applied. A general recommendation is 400 kg of Dolomite per acre for soils with a pH below 5.
- **Application Timing:** Dolomite should be applied during periods of soil moisture (onset or end of the rainy season).
- **Delay Rule:** Chemical fertilizers must not be applied simultaneously with Dolomite. A minimum waiting period of six weeks (1.5 months) is required after Dolomite application before inorganic NPK fertilizers can be applied.

3.2 Inorganic Fertilizer Recommendations

The standard inorganic fertilizer recommendation for cinnamon is an N:P2O5:K2O ratio of 23:7:15.

- **Mature Plants (Year 3 onwards, 4x3 ft spacing):** Apply 50g Urea, 25g Muriate of Potash (MOP), and 25g Eppawala Rock Phosphate (ERP) per bush annually.
- **Application Schedule:** The annual dosage should be split into two equal applications, timed with the onset of the Yala and Maha monsoon rains.
- **Placement:** For mature bushes, apply fertilizer in a circular band approximately 30 cm (12 inches) away from the base of the plant.

3.3 Integrated Plant Nutrition Management (IPNM)

To maintain soil health, IPNM combines inorganic and organic fertilizers.

- **Organic Integration:** Apply organic matter such as compost, Gliricidia leaves, or well-rotted cow dung. When using IPNM, the inorganic fertilizer dose can be reduced by 50%, supplemented with 1000g of organic manure per bush annually.
- **Biochar:** Biochar produced from cinnamon wood waste can be incorporated to improve soil carbon and nutrient retention.

4 Disease Management

The LLM should use these guidelines to help farmers identify and manage common cinnamon diseases.

4.1 Rough Bark Disease (Pestalotiopsis, Phomopsis, Colletotrichum spp.)

- **Symptoms:** Initially appears as small brown spots on immature green stems. These develop into larger brown patches with black margins, eventually causing the bark to become rough, dry, and cracked. It severely impacts bark quality and peelability.

- **Control:** Prune and burn infected branches. Ensure proper shade control and timely harvesting (every 4-6 months). Apply a 1% Bordeaux mixture or a fungicide containing Tebuconazole (10 ml per 10 L of water) to early-stage infections.

4.2 White Root Disease (Rigidoporus microporus)

- **Symptoms:** Leaves turn yellow and wilt, leading to plant death. White fungal threads (mycelium) are visible on the surface of the infected roots. Highly prevalent in old rubber lands or lands adjacent to rubber plantations.
- **Control:** Uproot and burn infected plants. Expose the remaining roots (4 inches deep) and apply a Tebuconazole drench (10 ml per 10 L water). Isolate the infected area by digging trenches. When planting in old rubber lands, apply 10g of Sulphur dust per planting hole one week prior to planting.

4.3 Brown Root Disease (Phellinus noxius)

- **Symptoms:** Similar to White Root Disease, but roots turn brown and have soil/sand adhering tightly to them. Distinct brown lines are visible beneath the bark.
- **Control:** Improve soil drainage, remove excess shade, and apply Tebuconazole drench as recommended for White Root Disease.

4.4 Leaf Blight (Colletotrichum spp.)

- **Symptoms:** Small brown blister-like spots on leaves that enlarge into circular necrotic patches. The necrotic tissue may fall out, leaving holes. Primarily affects nurseries and young plants during wet weather.
- **Control:** Remove shade, isolate infected plants, and apply a 1% Bordeaux mixture, Hexaconazole, or Tebuconazole.

5 Pest Management

5.1 Cinnamon Wood Borer / Pink Borer (Ichneumoniptera cinnamomumi)

- **Symptoms:** The larva bores into the base of mature stems, feeding on the tissue from the bark to the heartwood. Brown powdery frass is visible at the base of the plant. Infected stems may break and fall over.
- **Control:** The most effective control is earthing up (mounding soil) around the base of the plant to a height of about 2 inches, especially after the plant is 3 years old. If soil cannot be used, a clay/mud mixture can be applied. Ensure timely harvesting.

5.2 Thrips (Helionothrips annosus)

- **Symptoms:** Feeding causes the tips of young leaves to appear burnt and black. Shiny spots (excreta) are visible on the underside of leaves. Severe attacks can cause defoliation and dieback of shoots.
- **Control:** Pluck and destroy infected leaves early. For severe attacks, apply Imidacloprid (15 ml per 16 L water).

6 Harvesting and Processing (Post-Harvest Technology)

The LLM should advise on proper harvesting techniques to maintain the quality and safety of the final product.

6.1 Harvesting Indicators

- **Timing:** Harvesting begins 2-3 years after planting. The optimal time to harvest is when the plant flushes new leaves and they turn green, as this indicates sap flow, making peeling easier. Avoid harvesting during flowering, fruiting, or extreme dry periods.
- **Visual Cues:** The outer bark (subula) should be entirely brown, except for the very tips of the branches.
- **Peeling Test:** The bark must easily separate from the wood when tested with a knife.

6.2 Harvesting Technique

- **Cutting:** Cut the stem at a 45-degree angle, 1.5 to 2 inches above the ground level, using a sharp, clean knife. This promotes the outward growth of new shoots.
- **Selective Harvesting:** Leave at least one mature stem per bush unharvested (roughly 1 in 5 stems) to sustain the plant's nutritional needs and provide a future harvest.
- **Time of Day:** Harvest in the early morning. Prolonged exposure of cut stems to sunlight will cause them to dry out, making peeling difficult.

6.3 Processing (Peeling and Drying)

- **Scraping (Kurutta Sirima):** The outer brown bark must be completely scraped off using a specialized curved knife (Kokaththa) to reveal the yellow inner bark. Failure to remove the outer bark results in a lower-grade product with a poor taste, unattractive color, and higher susceptibility to insect damage.
- **Rubbing:** Rub the scraped stem with a brass rod to loosen the inner bark. Iron rods must not be used, as iron reacts with the cinnamon, turning the bark black.

- **Drying:** Freshly peeled bark must be air-dried in the shade on clean nylon nets or racks until the edges curl inward. Never expose the bark to direct sunlight. If using mechanical dryers, the temperature must not exceed 35°C to prevent the loss of volatile flavor compounds.
- **Quill Formation:** The standard length for a high-quality cinnamon quill is 42 inches.

6.4 Hygiene and Quality Standards

- **Sanitation:** Workers must wash hands, wear clean clothes, and refrain from eating, chewing betel, or spitting in the processing area.
- **Ground Contact:** Cinnamon stems and peeled bark must never touch the ground.
- **Standards:** Sri Lanka Cinnamon must adhere to SLS 81:2021 and ISO 6539:2014 standards. Moisture content must not exceed 14%. Sulphur dioxide limits must not exceed 150 mg/kg.